Is Anyone Home on Mars?

A rusty-red world stretches beneath a pale pink sky.

Dust covers the ground. Giant volcanoes rise above the land, deep valleys cut through the surface, and huge craters mark the rocky ground.

This is Mars, our neighbouring planet. It looks quiet and empty, but scientists have a big question:

Could anything be living beneath all that red dust?

In 1976, NASA sent a robot called Viking 1 to find out.

Meet Viking 1

Viking 1 wasn't built to drive across Mars. It was designed to stay in one place and study its surroundings.

Its cameras photographed the landscape. Other instruments measured the temperature, wind and air pressure.

But one piece of equipment had a particularly important job: a robotic arm.

At its end was a metal scoop that could reach down, dig into the ground and collect soil. The sample would then travel into a laboratory built inside the lander.

This was no ordinary laboratory. It had been designed to investigate whether Martian soil contained living organisms.

A Scoop of Martian Soil

On July 28, 1976, Viking's robotic arm collected its first soil sample.

The scoop scraped through the reddish ground, lifted the dirt and carried it back to the lander. Inside, a special system delivered the sample to the scientific instruments.

Scientists couldn't simply look at the soil and recognise a living microbe. They needed to see how it behaved under different conditions.

Viking carried several experiments to investigate this. Each one looked for a different sign of biological activity.

Did the Soil Respond to Food?

One experiment began by adding a nutrient-rich liquid to the soil.

It also contained radioactive carbon, which would help scientists detect certain gases if they were released.

The scientists had a prediction: if living microbes were present, they might use the nutrients and release gases that Viking could measure.

The instruments detected radioactive gas.

That was an interesting result. Something in the sample had reacted.

But there was another possibility. Chemicals in the Martian soil could produce a similar reaction without any living organisms.

The experiment had detected activity. It had not identified its cause.

When the Other Tests Gave Different Results

Viking had more than one way to examine the soil.

Another experiment exposed a sample to light and checked whether radioactive carbon dioxide became part of organic material. This could indicate that organisms were using carbon, as living things do.

A third experiment heated the soil and measured the gases released.

The results didn't give scientists a clear answer.

The biological tests showed some unexpected activity. But the chemical analysis found no detectable organic compounds at the levels the instrument could measure.

That was a problem. Scientists had signs of a possible reaction, but no convincing evidence that living microbes were responsible.

How Did Viking Work Through the Martian Nights?

All those experiments needed electricity. Viking also had to operate through freezing nights, when sunlight was unavailable.

So NASA equipped the lander with two special generators called radioisotope thermoelectric generators, or RTGs.

Inside each generator, radioactive plutonium produced heat as it naturally decayed. The generator converted some of that heat into electricity.

There were no solar panels on the lander. These generators supplied the power needed to operate its instruments throughout the mission.

A 90-Day Mission That Lasted Six Years

NASA had designed Viking 1 to operate for 90 days after landing.

It lasted much longer.

The lander continued working for more than six years. It photographed its surroundings, measured Martian weather and sent scientific results back to Earth.

Its robotic arm had collected soil. Its laboratory had carried out experiments. Its cameras had recorded a landscape no human had ever visited.

Then, in November 1982, communication with Viking 1 was lost.

A faulty command sent from Earth disrupted contact with the lander. Engineers tried to restore communication, but Viking never responded again.

Its cameras stopped transmitting. Its instruments stopped sending measurements. The robotic arm would never collect another sample.

The Equipment We Left Behind

Viking 1 still rests at its original landing site in Chryse Planitia, a broad plain on Mars.

Its mission is over, but its contribution to planetary science remains.

The lander showed scientists how to collect and examine soil on another planet. Its experiments also revealed how difficult it is to distinguish biological activity from ordinary chemical reactions.

Later Mars missions continued investigating the planet's rocks, soil and ancient environments for evidence that life may once have existed there.

Viking 1 never confirmed that Mars contained life. But it gave scientists valuable results from their first successful close-up investigation of Martian soil.

Today, the lander is still there. The instruments that once examined the red soil are no longer working, but the scientific questions they helped raise remain important.